Fuzzy Logic Assignment 2 - Theoretical

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1 Question 1 – Fuzzy sets

A fuzzy set A in X (classical set of objects, called the universe, whose generic elements are denoted x) is a set of ordered pairs:

$$A = \{(x, \mu_A(x)) | x \in X, \mu_A(x) \in [0, 1]\}.$$

The α-cut ($\alpha \in [0, 1]$) of a fuzzy set A is the ordinary set

$$A_\alpha = \{x \in X | \mu_A(x) \geq \alpha\}.$$

The strong α-cut ($\alpha \in [0, 1]$) of a fuzzy set A is the ordinary set

$$A_{\alpha^+} = \{x \in X | \mu_A(x) > \alpha\}.$$

The set of all distinct numbers in [0, 1] that are employed as membership grades of the elements of X in A is called the level set of A, denoted by L(A).

Answer the following:

a) Given any two fuzzy sets A and B, prove that the following properties hold:

$$(A \cup B)_\alpha = A_\alpha \cup B_\alpha, (A \cap B)_\alpha = A_\alpha \cap B_\alpha.$$

(Assume minimum and maximum operators for the intersection and union of fuzzy sets.)

Answer:
1.1.$\forall x \in (A \cup B)_\alpha$
$\Rightarrow max(A(x), B(x)) \geq \alpha$
$\Rightarrow A(x) \geq \alpha \vee B(x) \geq \alpha$
$\Rightarrow x \in A_\alpha \vee x \in B_\alpha$
$\Rightarrow x \in A_\alpha \cup B_\alpha$
$\Rightarrow (A \cup B)_\alpha \subseteq A_\alpha \cup B_\alpha$.

1.2.$\forall x \in A_\alpha \cup B_\alpha$
$\Rightarrow x \in A_\alpha \vee x \in B_\alpha$
$\Rightarrow A(x) \geq \alpha \vee B(x) \geq \alpha$
$\Rightarrow max(A(x), B(x)) \geq \alpha$
$\Rightarrow (A \cup B)(x) \geq \alpha$
$\Rightarrow x \in (A \cup B)_\alpha$
$\Rightarrow A_\alpha \cup B_\alpha \subseteq (A \cup B)_\alpha$.

1.3.From 1.1 and 1.2 $\Rightarrow (A \cup B)_\alpha = A_\alpha \cup B_\alpha$

2.1.$\forall x \in (A \cap B)_\alpha$
$\Rightarrow (A \cap B)(x) \geq \alpha$
$\Rightarrow min(A(x), B(x)) \geq \alpha$
$\Rightarrow A(x) \geq \alpha, B(x) \geq \alpha$
$\Rightarrow x \in A_\alpha, x \in B_\alpha$
$\Rightarrow x \in A_\alpha \cap B_\alpha$

$\Rightarrow (A \cap B)_\alpha \subseteq A_\alpha \cap B_\alpha$.

2.2.$\forall x \in A_\alpha \cap B_\alpha$
$\Rightarrow x \in A_\alpha, x \in B_\alpha$
$\Rightarrow A(x) \geq \alpha, B(x) \geq \alpha$
$\Rightarrow min(A(x), B(x)) \geq \alpha$
$\Rightarrow (A \cap B) \geq \alpha$
$\Rightarrow x \in (A \cap B)_\alpha$
$\Rightarrow A_\alpha \cap B_\alpha \subseteq (A \cap B)_\alpha$.

2.3.From 2.1 and 2.2 $\Rightarrow (A \cap B)_\alpha = A_\alpha \cap B_\alpha$.

b) The support of A, denoted *supp(A)*, is defined as the set of elements of X that have nonzero membership in A. The core of A, denoted *core(A)*, is defined as the set of elements of X that have membership in A equals to 1.

How do *supp(A)* and *core(A)* relate to the α-cuts and the strong α-cuts of A?

Answer:
Hypothesis:
α-cut: $A_\alpha = \{x \in X | \mu_A(x) \geq \alpha\}$
Strong α-cut: $A_{\alpha^+} = \{x \in X | \mu_A(x) > \alpha\}$
Support: $supp(A) = \{x \in X | \mu_A(x) > 0\}$
Core: $core(A) = \{x \in X | \mu_A(x) = 1\}$

Observations:
For $\alpha = 1$:
$A_1 = \{x \in X | \mu_A(x) \geq 1\} = \{x \in X | \mu_A(x) = 1\} = core(A)$
For $\alpha = 0$:
$A_{0^+} = \{x \in X | \mu_A(x) > 0\} = supp(A)$

Conclusions:
The α-cut of set A when $\alpha = 1$ is equal to the core of A.
The Strong α-cut of set A when $\alpha = 0$ is equal to the support of A.

c) Given the discrete fuzzy sets A=$\{(0.2, x_1), (0.4, x_2), (0.6, x_3), (0.8, x_4), (1, x_5)\}$, obtain L(A), and provide all the distinct $\alpha - cuts$ of A.

What is the relationship between A_{α_1} and A_{α_2} when $\alpha_1 < \alpha_2$?

Answer:

$$L(A) = \{0.2, 0.4, 0.6, 0.8, 1\}$$
$$A_{0.2} = \{x_1, x_2, x_3, x_4, x_5\}$$
$$A_{0.4} = \{x_2, x_3, x_4, x_5\}$$
$$A_{0.6} = \{x_3, x_4, x_5\}$$
$$A_{0.8} = \{x_4, x_5\}$$
$$A_1 = \{x_5\}$$

In the exemple above, $\forall \alpha_1, \alpha_2 \in L(A), \alpha_1 < \alpha_2 \Rightarrow A_{\alpha_2} \subset A_{\alpha_1}$

Generalization:
Let A a fuzzy set in X with the level set L(A) and $\alpha_1, \alpha_2 \in L(A), \alpha_1 < \alpha_2$.
Using Reductio Ad Absurdum principle we will consider $A_{\alpha_2} \not\subset A_{\alpha_1}$
$\Rightarrow \exists x \in A_{\alpha_2}$ as $x \notin A_{\alpha_1}$
$\Rightarrow \mu_A(x) \geq \alpha_2, \mu_A(x) < \alpha_1$
$\Rightarrow \alpha_1 > \mu_A(x) \geq \alpha_2$
$\Rightarrow \alpha_1 > \alpha_2$ - contradiction
$\Rightarrow A_{\alpha_2} \subset A_{\alpha_1}$

d) The membership function of A can be expressed in terms of the characteristic functions of its $\alpha - cuts$ according to the formula:

$$\mu_A = sup_{\alpha \in [0,1]} \alpha \cdot \mu_{A_\alpha}(x) \text{ where } \mu_{A_\alpha}(x) = \begin{cases} 1 & iff \quad x \in A_\alpha \\ 0 & otherwise \end{cases}$$

In the case of a discrete fuzzy set we have $\alpha \in L(A)$. Show that this is true for the discrete fuzzy set given in c).

Answer:

$$L(A) = \{0.2, 0.4, 0.6, 0.8, 1\}$$

$sup_{\alpha \in L(A)} \alpha \cdot \mu_{A_\alpha}(x) = max\{0.2 \cdot \mu_{A_{0.2}}(x), 0.4 \cdot \mu_{A_{0.4}}(x), 0.6 \cdot \mu_{A_{0.6}}(x), 0.8 \cdot \mu_{A_{0.8}}(x), 1 \cdot \mu_{A_1}(x)\}$
For $x = x_1$
$max\{0.2 \cdot \mu_{A_{0.2}}(x_1), 0.4 \cdot \mu_{A_{0.4}}(x_1), 0.6 \cdot \mu_{A_{0.6}}(x_1), 0.8 \cdot \mu_{A_{0.8}}(x_1), 1 \cdot \mu_{A_1}(x_1)\} = 0.2 = \mu_A(x_1)$
For $x = x_2$
$max\{0.2 \cdot \mu_{A_{0.2}}(x_2), 0.4 \cdot \mu_{A_{0.4}}(x_2), 0.6 \cdot \mu_{A_{0.6}}(x_2), 0.8 \cdot \mu_{A_{0.8}}(x_2), 1 \cdot \mu_{A_1}(x_2)\} = max\{0.2, 0.4\} = 0.4 = \mu_A(x_2)$
For $x = x_3$
$max\{0.2 \cdot \mu_{A_{0.2}}(x_3), 0.4 \cdot \mu_{A_{0.4}}(x_3), 0.6 \cdot \mu_{A_{0.6}}(x_3), 0.8 \cdot \mu_{A_{0.8}}(x_3), 1 \cdot \mu_{A_1}(x_3)\} = max\{0.2, 0.4, 0.6\} = 0.6 = \mu_A(x_3)$
For $x = x_4$
$max\{0.2 \cdot \mu_{A_{0.2}}(x_4), 0.4 \cdot \mu_{A_{0.4}}(x_4), 0.6 \cdot \mu_{A_{0.6}}(x_4), 0.8 \cdot \mu_{A_{0.8}}(x_4), 1 \cdot \mu_{A_1}(x_4)\} = max\{0.2, 0.4, 0.6, 0.8\} = 0.8 = \mu_A(x_4)$
For $x = x_5$
$max\{0.2 \cdot \mu_{A_{0.2}}(x_5), 0.4 \cdot \mu_{A_{0.4}}(x_5), 0.6 \cdot \mu_{A_{0.6}}(x_5), 0.8 \cdot \mu_{A_{0.8}}(x_5), 1 \cdot \mu_{A_1}(x_5)\} = max\{0.2, 0.4, 0.6, 0.8, 1\} = 1 = \mu_A(x_5)$
$\Rightarrow sup_{\alpha \in L(A)} \alpha \cdot \mu_{A_\alpha} = \mu_A$

2 Question 2 – Decision making in a fuzzy environment

Fuzziness can be introduced at several points in the existing models of decision making.

a) Bellman and Zadeh in 1970 suggested a fuzzy model of decisions that must accommodate certain constraints C and goals G. Provide a description of this model.

Answer:
In their paper, "Decision Making in a Fuzzy Environment" Bellman and Zadeh present fuzzy goals and constraints as fuzzy sets, like in the following examples: "the result x should be a bit larger than 10" as fuzzy goal and "the value should be around between 5 and 6" as a fuzzy constraint. Fuzziness is determined in both sets by the words as "a bit" or "around".

Following the standard path of a decision process which covers: a set of alternatives, a set of constraints for decision and a method to evaluate the performance of the decision, the authors propose a model of decision in fuzzy environments.

For a set of alternatives X, each fuzzy goal G and fuzzy constraint C will be a fuzzy set as exemplified above, among the adjacent membership function (i.e. μ_G and μ_c). While it might be obvious that the set of alternatives and constraints have the same role in the fuzzy environment and standard decisions, the fuzzy goals are a slightly different form of the performance evaluation method. Using goals as fuzzy sets helps us to treat the goals and constraints in the same manner.

Since the set of alternatives must respect the goals and constraints, it means they will be measured under the intersection of these ($G \cap C$) while the membership function will be the conjunction the all the membership functions for C and G.

According to this model, a decision in the fuzzy environment will be a fuzzy set generated by the previously mentioned intersection, i.e. $D = G \cap C$ with the membership function $\mu_D = \mu_G \wedge \mu_C$. In case that goals and constraints have different importance, the membership function can be computed as a convex combination of them, multiplied by coefficients which will determine the importance.

The *optimal decision* (i.e. μ_{D^m}) is represented in this model by the alternatives which have the maximal value for the membership function, i.e. $\mu_{D^m} = \max \mu_D$. [1]

b) Suppose we must choose one of four different jobs a, b, c, and d, the salaries of which are given by the function f such that:

f(a) = 30,000, f(b) = 25,000, f(c) = 20,000 and f(d) = 15,000.

Our goal is to choose the job that will give us a high salary given the constraints that the job is interesting and within close driving distance.

The first constraint of interest value is represented by the fuzzy set

$$C_1 = \{(0.4, a), (0.6, b), (0.8, c), (0.6, d)\}.$$

The second constraint concerning the driving distance to each job is defined by the fuzzy set

$$C_2 = \{(0.1, a), (0.9, b), (0.7, c), (1, d)\}.$$

The fuzzy goal G of a high salary is defined by the membership function

$$\mu_G(x) = \begin{cases} 0 & for \quad x < 13,000 \\ -0.00125(\frac{x}{1000} - 40)^2 + 1 & for \quad 13,000 \leq x \leq 40,00 \\ 1 & for \quad x > 40,000 \end{cases}$$

Which is the best job when applying Bellman and Zadeh's fuzzy decision model?

Answer:

The membership functions of the constraints is expressed in the following table:

x	a	b	c	d
μ_{C_1}	0.4	0.6	0.8	0.6
μ_{C_2}	0.1	0.9	0.7	1

The fuzzy goal G_s of a job having a high salary can be expressed as $\mu_{G_s} = \mu_G \circ f$.

$$\mu_{G_s}(a) = \mu_G(f(a)) = \mu_G(30000) = -0.00125(\frac{30000}{1000} - 40)^2 + 1 = 0.875$$

$$\mu_{G_s}(b) = \mu_G(f(b)) = \mu_G(25000) = -0.00125(\frac{25000}{1000} - 40)^2 + 1 = 0.71875$$

$$\mu_{G_s}(c) = \mu_G(f(c)) = \mu_G(20000) = -0.00125(\frac{20000}{1000} - 40)^2 + 1 = 0.5$$

$$\mu_{G_s}(d) = \mu_G(f(d)) = \mu_G(15000) = -0.00125(\frac{15000}{1000} - 40)^2 + 1 = 0.21875$$

The membership function of the goal is expressed below:

x	a	b	c	d
μ_{G_s}	0.875	0.71875	0.5	0.21875

The decision fuzzy set is the intersection of the goals and constraints, while the membership function of the decision is the conjunction of their membership functions, i.e.

$$D = C_1 \cap C_2 \cap G_s$$

$$\mu_D = \mu_{C_1} \wedge \mu_{C_2} \wedge \mu_{G_s}$$

$$\mu_D = min\{\mu_{C_1}, \mu_{C_2}, \mu_{G_s}\}$$

The membership function and fuzzy set of the decision are expressed below:

x	a	b	c	d
μ_D	0.1	0.6	0.5	0.21875

$$D = \{(0.1, a), (0.6, b), (0.5, c), (0.21875, d)\}$$

The best job is the one with the maximal value of the membership function μ_{D^m}(x) = max μ_D(x) = 0.6, which is b.

References

[1] BELLMAN, R.E and ZADEH, L.A. (1970) Decision making in a fuzzy environment. *Management Science*, 17 (1), pp. 141-164.

3 Question 3 - ANFIS

Consider the IRIS data seen in the practical exercises. Explain how data such as this can be used to generate a fuzzy system using ANFIS. Illustrate your answer with examples from the lab work you did using ANFIS with the IRIS data. Include in your answer a brief discussion of when it is appropriate to use ANFIS to build a fuzzy system and identify some of the decisions that need to be made in order to generate a useful system.

Answer:

ANFIS are a particular type of Fuzzy Inference Systems which will compute the membership functions instead of having it specified when it is designed. ANFIS (Adaptive Neuro-Fuzzy Inference System) use a similar approach to neural networks using sets of inputs and outputs to generate a structure that can correlate to these sets. For ANFIS, this structure is represented by the estimated membership function.

ANFIS are very useful when we have large datasets of inputs and expected outputs for a specific model. Sometimes in these, cases the membership function is hard to be estimated by simple rules. Also, in some situations, the model does not have clear characteristics to be perceived by human. It is easier to delegate this task to the ANFIS which will lean rules to satisfy the input/output datasets. In fact, ANFIS compute mathematical relationships to link inputs to outputs (i.e. backpropagation method or a combination of least-squares and backpropagation gradient descent method [1]).

To generate an useful ANFIS, we need to consider the following aspects:

- As for neural networks, it is useful that a part of the datasets is used to train the model and estimate the MF (membership function) and another part for testing purpose. If no testing set is used, overfitting might occur.
- Using validation data (i.e. a third part of the initial input/output dataset) we can test the performance of the ANFIS after each epoch on different data than it was trained. This method is practical when we have large and noisy datasets because they will often tend to overfit [1].
- Consistent sets of input and output are necessary. Small datasets will not provide enough examples for the ANFIS to learn the rules.
- Using gradient descent method to compute the parameters, specific aspects of this procedure must be considered. If the initial learning rate is too big, we might miss the local minimum and not obtain a small enough error. Increase or decrease rate of the step size (learning rate) must take this aspect into account too.
- Depending on the number of parameters, more training epochs might be necessary.

IRIS data is a benchmark dataset of 150 5-dimensional samples which has a classification task as goal. It contains three classes of flowers which are the 1-dimensional output set expected by ANFIS, i.e.

$$O = \{1, 2, 3\}.$$

The input set expected by ANFIS is a 4-dimensional one, having 4 characteristic for each sample, i.e.

$$I = \{(x_1, x_2, x_3, x_4) | x_1 \in [4.3, 7.9], x_2 \in [2, 4.4], x_3 \in [1, 6.9], x_4 \in [0.1, 2.5]\}.$$

For the training set, 125 samples will be selected, while for the testing one, 25 samples.

```
%The already prepared sets of train and test data must be loaded:
load iristrain.dat;
load iristest.dat;
```

To create an ANFIS, the first step is to define a number of membership functions for each variable and the types of the membership functions. The "genfis1" command is used to generate the ANFIS, specifying the train dataset, the number of membership functions and their type.

```
nr_mf = 3;
mf = 'trimf';
irisfis = genfis1(iristrain,nr_mf,mf);

writefis(irisfis,'irisfis.m'); % Use this to store the ANFIS
fuzzy irisfis; % Use to visualize the ANFIS
```

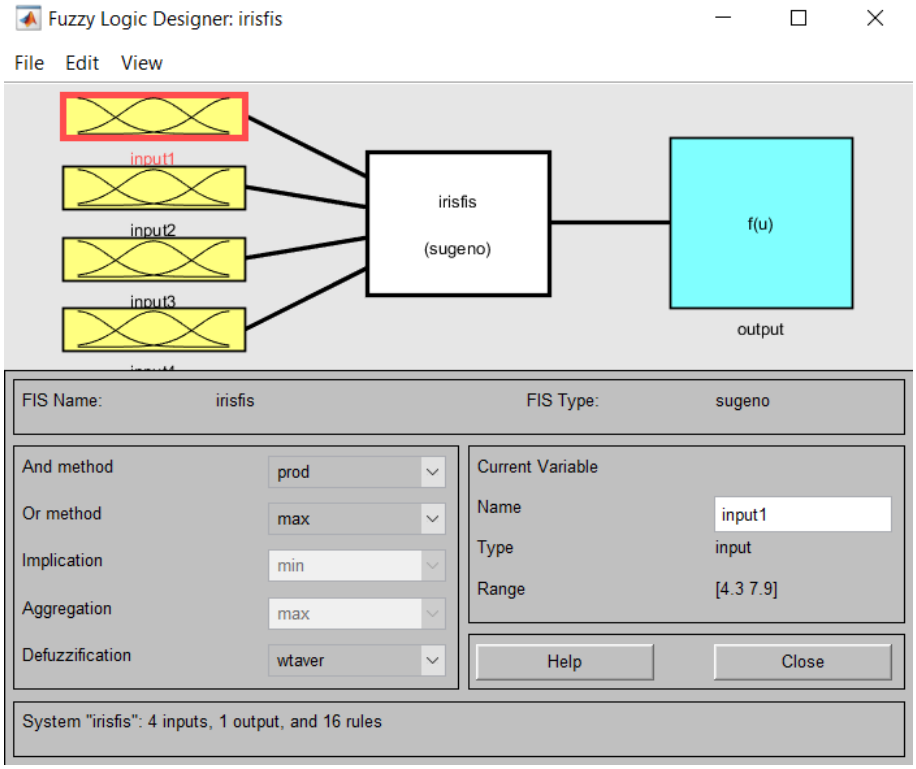


Figure 1: ANFIS Viewer

After generating the ANFIS, we have to train it the way a neural network is trained. This can be done using the "anfis" command and specifying the train dataset. This will tune the FIS parameters using the expected input and output from the dataset. The FIS structure can also be specified as a starting point in the second parameters of this function (e.g. in our example it is irisfis).

For specifying the training options, a third parameter can be used. This parameter is an 1-D array of 5 elements as it follows:

- Number of epochs;
- Goal for the error;
- Initial learning rate (i.e. step size of gradient descent algorithm used by ANFIS to compute parameters);
- Decrease rate for the step size;
- Increase rate of the step size.

  If options are not specified, default values will be used by Matlab. When setting both number of epochs and error goal, the training will stop when the first of them will be reached.

```
%Specify train dataset, FIS, number of training epochs and error goal
nr_epochs = 50;
err_goal = 0.0001;
options = [nr_epochs, err_goal];
irisfis = anfis(iristrain, irisfis, options);
```

After training, Matlab will print the information regarding the trained ANFIS and the error obtained on each epoch.

Example:

```
ANFIS info:
        Number of nodes: 193
        Number of linear parameters: 405
        Number of nonlinear parameters: 36
        Total number of parameters: 441
        Number of training data pairs: 125
        Number of checking data pairs: 0
```

```
        Number of fuzzy rules: 81

Error per epoch:
26        0.00010553
27        0.000100571
28        9.87149e-05

Error goal (0.0001) reached --> ANFIS training completed at epoch 28.
```

To evaluate the ANFIS, the "evalfis" command can be used. This will simulate the specified FIS for an input dataset and return the assumed output. The error is the difference between the output of the FIS and the initial output known.

```
input_test_dataset = iristest(:,1:4);
output_test_dataset = iristest(:,5);
y = evalfis(input_test_dataset,irisfis);
error = y - output_test_dataset;
```

To visualize the results, we can plot the error percentage or the expected results against the obtained one. The error percentage is the report between the obtained error and the expected output. Both graphics can be plotted as it follows:

```
figure(1);
percentage_error = error/output_test_dataset;
subplot(2,1,1);
plot(percentage_error)

subplot(2,1,2);
plot(y, output_test_dataset, '*');
```

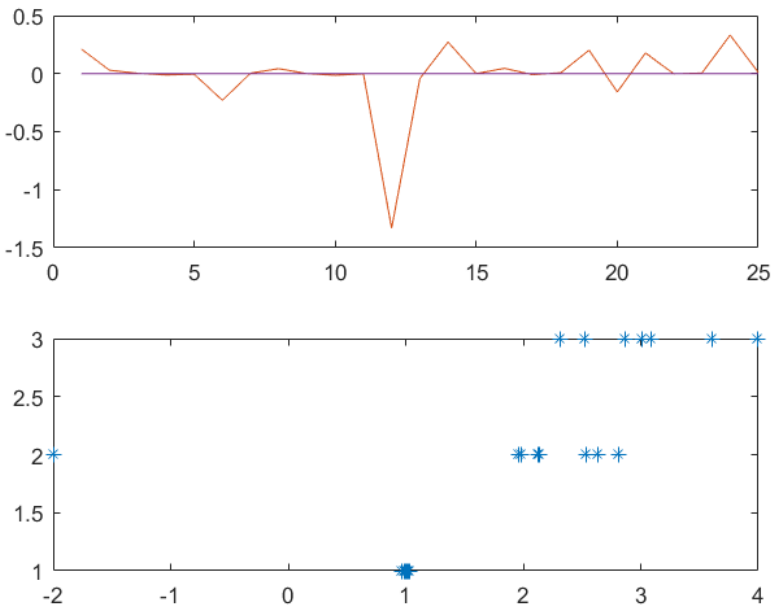


Figure 2: Graphic results

References

[1] MATHWORKS, INC (2018) *Fuzzy Logic Toolbox^{TM} User's Guide.* [Online] pp. 147-155. Available from: *https://www.mathworks.com/help/pdf$_d$oc/fuzzy/fuzzy.pdf*. [Accessed 30/12/18].

4 Question 4 – Type 2 fuzzy logic

Write a short essay (about 2 pages not including bibliography & references) to discuss why there is a perceived need for the type-2 fuzzy logic and consider the areas of application where type-2 fuzzy sets might provide a solution. Your answer should include a brief definition of a type-2 fuzzy set (illustrate with diagram(s) and example(s)). It should also make reference to issues associated with the practical implementation of type-2 fuzzy logic and to the approaches that have been developed.

Answer:

Importance of type 2 fuzzy logic

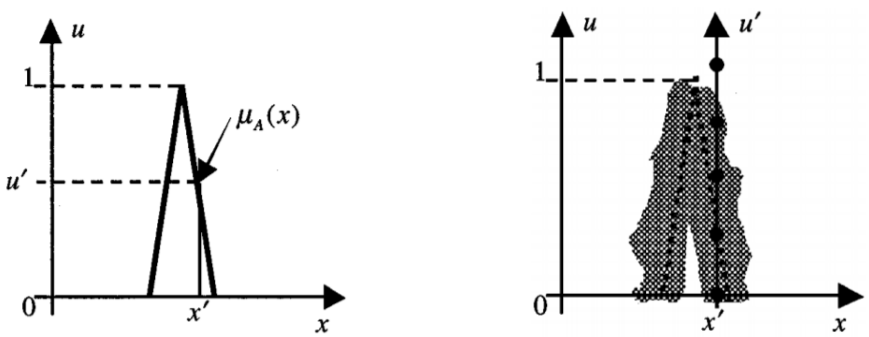
Abstract

To model the subjective situations, a human takes a set of decisions, according to the imposed rules. A mathematical model of this decisions was introduced by fuzzy set theory, offering systems the capacity to take human-like decisions. However, decisions in real world implies more fuzzification and variables than type-1 fuzzy logics can handle. This limitation leads to the necessity to increase the uncertainty level when dealing with fuzzy sets and to the development of fuzzy-2 logic.

Introductive notions:

Membership function in type-1 fuzzy logic provides for each input x a single value y $\in [0, 1]$. An example for this scenarios is that, considering a specific temperature of $5°C$, we can say it is 0.8 quite cold in type-1 fuzzy logic. Considering we want to characterize this temperature as quite cold for different perspectives, we will find ourself limited. Using type-2 fuzzy logics we will be able to say that the value is quite cold in the interval [0.5,1].

Generalising this concept, considering the type-1 membership function for a variable, in figure 3a it is noticeable that for the value $x' \in A$ we have $\mu_A(x') = u'$ where $u' \in [0, 1]$. If the points in the graphical representation are shifted in different directions, then an apparently blurred region will be created as in figure 3b. In this situation, for the input x' we will no longer have just one possible value of the membership function, but a set of possible values increasing the dimension of the membership function.



(a) Type-1 MF (b) Type-2 MF

Figure 3: Graphical membership function [12]

This observation leads us to a formal definition of type-2 fuzzy sets.

Definition 1: A type-2 fuzzy set is determined by a type-1 membership function μ_A, a type-2 membership function $\mu_{\tilde{A}}$ and an input variable x according the following formula:

$$\tilde{A} = \{((x, u), \mu_{\tilde{A}}(x, u)) | \forall x \in X, \forall u \in J_x \subseteq [0, 1]\}[12]$$

$$0 \leq \mu_{\tilde{A}}(x, u) \leq 1[12]$$

$\tilde{A}$ is an union for the continuous case,

$$\tilde{A} = \int_{x \in X} \int_{u \in J_x} \mu_{\tilde{A}}(x, u)/(x, u), J_x \subseteq [0, 1][12]$$

and a sum for the discrete case,

$$\tilde{A} = \sum_{x \in X} \sum_{u \in J_x} \mu_{\tilde{A}}(x, u)/(x, u), J_x \subseteq [0, 1].$$

Definition 2: For a particular x' value of a variable $x \in X$ we have type-2 membership function $\mu_{\tilde{A}}(x = x', u) \equiv \mu_{\tilde{A}}(x')$ called *vertical slice.*

Considering the values $u \in U$ of a type-1 membership function, another type-1 membership function $f_{x'}$, $0 \leq f_{x'}(u) \leq 1$ with domain on U, a variable $x \in X$ and $J_x \in [0, 1]$ a set of possible values of u, the type-2 membership function is defined after the following formula:

$$\mu_{\tilde{A}}(x') = \int_{u \in J_{x'}} f_{x'}(u)/u.[12]$$

Considering this, the type-2 fuzzy set $\tilde{A}$ can be written as:

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) | \forall x \in X\}$$

Example: A type-2 fuzzy set can modellated using the following elements:

- A set of variables $X = \{1, 2\}$
- The values of the type-1 membership function $U = \{0, 0.8\}$
- The possible values $J_1 = \{0\}, J_2 = \{0, 0.8\}$
- A type-2 membership function $\mu_{\tilde{A}}$
  - $\mu_{\tilde{A}}(1) = 0.3/0$
  - $\mu_{\tilde{A}}(2) = 0.35/0 + 0.6/0.8$

Example: Another graphical representation of a type-2 membership function is the Footprint of Uncertainty from figure 4. In this figure we have $x \in X = \{1, 2, 3, 4, 5\}$ and the values of a type-1 membership function $\mu_A, u \in U = \{0.2, 0.4, 0.5, 0.6, 0.8\}$ in the initial 2D plan. The type-2 membership function $\mu_{\tilde{A}} \in \tilde{U}$ generates the 3D plan.

This graphical representation offers a better understanding of the 3-D representation of a fuzzy set letting us visualize in 2-D the shape of a type-2 membership function.

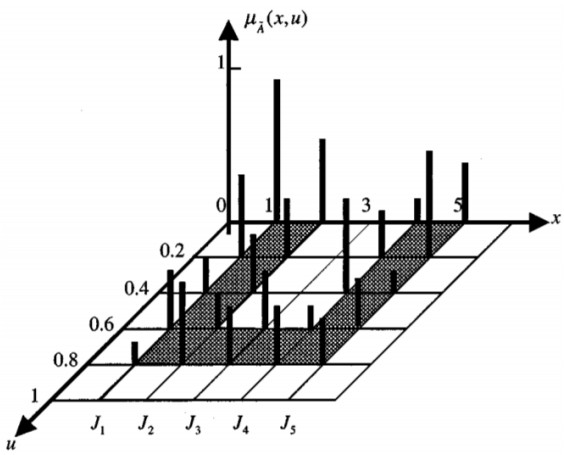


Figure 4: Footprint of Uncertainty [12]

A vertical slice of the type-2 membership function can be observed for any value of $x \in X$, for example $x = 1$. We will notice that $J_1 = \{0, 0.2, 0.4, 0.6, 0.8\}$. According the formula from Definition 2 we must identify the values for $f_{x=1}$, i.e. $f_1(0) = 1, f_1(0.2) = 0.5, f_1(0.4) = 0.2, f_1(0.6) = 0.35, f_1(0.8) = 0$. It means that we can write the type-2 membership function $\mu_{\tilde{A}}(1) = 1/0 + 0.5/0.2 + 0.2/0.4 + 0.35/0.6 + 0/0.8$.

Usage

The principal reason to use type-2 fuzzy is that in some situations, the crisp membership function used for type-1 fuzzy sets can not model more complex situation where a fuzzy membership function is necessary.

A good example in this direction is represented in the engineering field. In his paper, Hagras [3] presents the importance of a higher uncertainty level of type-2 fuzzy sets for a new generation of fuzzy controllers. Also, he highlights the applicability of these sets in fields as: industrial, mobile robots and intelligent systems control.

Type-2 fuzzy sets proved themselves useful when it comes to natural language processing. Descriptive linguistic terms for an uncertain scenario can be perceived different from a person to another. Subjective human perception introduces a high level of ambiguity. A simple but clear example in this direction is the human perception over the warmness in certain environment conditions. Consider three different persons working in the same office i.e. $\{person_1, person_2, person_3\}$ and the temperature being set to $20°C$. Due different reasons as health or physical condition, $person_1$ will feel cold in this condition, $person_2$ will feel comfortable and $person_3$ will consider it is too hot. If they will be asked to approximate on which level it is rather warm they could provide the following set of answers $\{(person_1, 0.1), (person_2, 0.7), (person_3, 1)\}$. All answers are correct and can be well modelled using a type-2 fuzzy system. In their paper [4], Wagner and Hagra present a novel approach to design a type-2 fuzzy set for the subjective human perception of linguistic terms. They use linguistic labels and create surveys for participant to record and process them in type-2 fuzzy sets.

The natural variation of certain values is easily perceived by human mind but hard to mathematically model. Sensor measurements of physical devices generate a high level of uncertainty due the environment impact. The sensors can be affected by any environmental change and a clear membership function cannot be used due to alterations of the initial conditions. In their paper [4], Wagner and Hagra present a method to model a type-2 fuzzy set for a sonar sensor. Their approach is applicable for any device responsible for measurement using sensors.

A wide rage of application of type-2 fuzzy sets is presented by Ahmad in his paper [2]. Many systems can vary in time, but the way they will vary in time is not know or cannot be predicted. In this situation, a fuzzy membership function to model the output is necessary as in mobile communication.

Pattern recognition tasks are widely used for multiple subjects. In [5] Jia and Liu present the importance of type-2 fuzzy sets in this field. Pattern recognition presents a large set of uncertainties generated by their goal to identify the pattern. We have to extract features and partition them according the rules, choose hypotheses and design an algorithm. But features are uncertain, so are the hypothesis, leading to uncertain modelling of the algorithm. Usually we assume the parameters are random variables, but using type-2 fuzzy sets we can take into account all possible uncertain parameters. Also, noisy training data has the same effect of introducing uncertainty.

Issues

Performance: As we can notice in the first two definitions, the mathematical representation of type-2 fuzzy sets is getting more and more complicated. In their paper, Mendel and John [12] present a simple but yet mathematically complex approach to understand type-2 fuzzy set operation. The more complex is the mathematical representation of a structure, the more computationally expensive it will be. In practice this is the biggest issue related to this fuzzy sets.

While type-1 fuzzy sets allows us to implement and compute them on regular computers, type-2 fuzzy sets requires more performant devices, meaning better hardware component and more time for computations. This hardware limitation is intended to be overcome, different solution being provided in studies as [8], [9], [10].

System modelling: Creating a type-2 fuzzy system, as for type-1 fuzzy systems requires expert analysis of the model that needs to be structured. Type-1 membership functions are sometimes hard to interpret by experts, but for type-2 fuzzy system this problem increases in difficulty. The rules, the shape of a fuzzy set, the inferencing operators are all parameters that need to be identified by experts and increasing the dimension of their interpretation can generate issues for a proper system modelling.

Type-2 membership functions Figure 3 illustrates an abstract interpretation of type-2 membership function 3b, starting from a well defined type-1 membership function 3a. Even if this abstract presentation of a type-2 MF seems logical, this is not offering a clear mathematical interpretation of this kind of function, but more likely an ambiguous explanation. The points in figure 3a can be blurred in any direction and here it arises a set of questions such as: *In which direction is appropriate to do this for best result?; How we define the best solution?*. Depending on how we interpret the fuzzy form of the membership function we can obtain very different results. Blurring is not the only way to create type-2 MF, this introducing ambiguity in comparison to type-1 MF where we have clear function (i.e. gaussian MF, trapezoidal MF, etc.).

ANFIS When large datasets of inputs and outputs are available, ANFIS are a great solution. Theoretically, it is possible to compute type-2 membership function for a fuzzy system, according to studies as [7] [11], methods to tune the learning parameters being already determined and documented [1]. Practically, if we consider that performance is a limitation for computing standard type-2 fuzzy systems, then the performance to compute the type-2 MF with ANFIS will drastically be increased. It is hard to estimate how efficient is a system and what is the best trade off between complexity and costs.

Defuzzification: Building a type-2 fuzzy system we model different uncertainty levesl. When we defuzzify the output of such systems, we will reduce it to a crisp number without being able to capture more dimensional information. This leads to information loss from the fuzzy system to the output [6].

Conclusion Type-2 fuzzy sets brings up a new approach for modelling uncertainty, being a great solution in multiple contexts and overcoming the type-1 fuzzy sets limitations. Nevertheless, they generate a series of issues which can determine us not to use them. Researchers in this field are constantly looking for solutions to improve their usability, but using type-2 fuzzy sets currently implies a trade off between computation complexity and theoretical performance.

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